

Multiple Choice (10 marks)

1. Message authentication is a service beyond
 - (a) Message Confidentiality
 - (b) Message Integrity
 - (c) Message Splashing
 - (d) Message Sending
2. _____ is also a part of darknet that is employed for transferring files anonymously.
 - (a) Freenet
 - (b) ARPANET
 - (c) Stuxnet
 - (d) Internet
3. Encryption and decryption provide secrecy, or confidentiality, but not
 - (a) Authentication
 - (b) Integrity
 - (c) Privacy
 - (d) All of the above
4. Encapsulating Security Payload (ESP) belongs to which Internet Security Protocol?
 - (a) Secure Socket Layer Protocol
 - (b) Secure IP Protocol
 - (c) Secure Http Protocol
 - (d) Transport Layer Security Protocol
5. A hash function guarantees the integrity of a message. It guarantees that the message has not be
 - (a) Replaced
 - (b) Overview
 - (c) Changed
 - (d) Violated

True / False (10 marks)

Answer True or False

6. True or False: A buffer overflow occurs when a pointer accesses memory not allocated to it, leading to potential data corruption or security vulnerabilities.
7. True or False: Only biometric methods are considered secure forms of authentication in modern cybersecurity practices.
8. True or False: Correctness is widely recognized as a fundamental security property alongside confidentiality and integrity.
9. True or False: The AH Protocol ensures both data integrity and privacy for the packets it protects during transmission.
10. True or False: A packet filter firewall functions at the application or transport layer, evaluating data packets for specific application protocols.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to round every number of a given list of numbers and print the total sum multiplied by the length of the list.
12. Write a python function to convert the given string to lower case.

End of Paper